

Contact

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Top Skills

Interdisciplinary Collaboration

Graphic Design

Deconvolution

Languages

French (Elementary)

Spanish (Native or Bilingual)

English (Native or Bilingual)

Publications

Venezuelan Immigration and
Integration to Trinidad and Tobago in
Recent Years

Santiago Aguilera

Perkiomen School '27 | Aspiring Physician-Scientist | Biomedical
Research | Global Health Advocate | Student Leader
Pennsburg, Pennsylvania, United States

Summary

Student at Perkiomen School aspiring to become a physician-scientist focused on biomedical innovation and advocating for global health. My interests revolve around clinical medicine, computational biology, medical research, and healthcare technology, with a mission on improving access to care for underserved populations. Beyond the classroom, I have conducted independent research, founded my school's HOSA chapter, volunteered and shadowed physicians in hospital settings, and continue to explore how conducting research and developing applications and websites are key in enhancing patient care and medical decision-making. I enjoy collaborating with researchers, healthcare professionals, and students who share a passion for advancing medicine through science, technology, and service.

Experience

Collaborative Summer Research eXperience (CSR-x)

Research Mentee

July 2026 - August 2026 (2 months)

Participated in a highly selective, 5-week remote student-led research program as member of a cohort of biomedical science studies in researching human exposure to polycyclic aromatic hydrocarbons and their association with epigenetic aging, DNA methylation, and lung cancer development.

Hospital Regional de Alta Especialidad Veracruz

Pediatric Surgeon Observer

July 2026 - August 2026 (2 months)

Veracruz, VER

Shadowed pediatric / neonatal surgeons, professionals, and residents in a public, government-funded pediatric hospital Spanish-speaking environment for 60+ hours. Had the opportunity to work with and learn from health professionals in a variety of fields ranging from pediatrics, surgery,

traumatology, orthopedics, anesthesiology, nursing, immunology, allergology, and cardiology.

In this environment, I collaborated with professionals in communicating with families about healthcare necessities while being present in various surgical procedures and meetings with the doctor I shadowed.

Dedicated time outside of the hospital in developing a website to refine digital and prescription systems for families without medical insurance and opportunities to optimize healthcare to a greater degree.

GVMUN

Journalism Intern

March 2026 - August 2026 (6 months)

United States

Worked on GVMUN as a Journalism Intern, this experience helped me enhance skills in writing and managing time efficiently when exposed with deadlines to meet during periods of publication. It encouraged me to learn further about necessary knowledge surrounding current world events at the time of employment and helped me attain greater proficiency in international relations and journalism.

Carnegie Mellon University

Student Researcher

June 2026 - July 2026 (2 months)

Pittsburgh, PA

Participated in an intensive, selective, 4-week program on computational biology as a student in Carnegie Mellon University with sessions in coding, laboratory work, research, and algorithm development with support from college professors and teaching assistants.

Finalist in final research and algorithm project. Competed against other groups of students in designing and incorporating algorithms addressing real-life issues. My group focused on planning and developing cell deconvolution and prognosis algorithms targeted towards the Glioblastoma tumor in a website. Collaborative research on the microenvironment, Glioblastoma, and computational methods was conducted in which algorithms were constructed through bulk RNA-seq of data from TCGA-GBM from 153 available samples to answer whether cell composition from the NNLS/SVR deconvolution algorithm was a major factor in Glioblastoma prognosis.

Phillips Exeter Academy

Summer Student

July 2025 - August 2025 (2 months)

Exeter, NH

Participated in Phillips Exeter Academy's Summer Program as a member of the selective Charles J. Hamm '55 Leadership Program. Focusing on social justice and community engagement, we focused on the philosophy, the interpretation and execution of leadership. In this program, students attended the Edward M. Kennedy Institute, and worked on various projects to encourage students in taking action and being leaders.

Courses included the practical leadership seminar and leadership analysis. Projects in these classes led to the creation of physical models to represent and highlight the impact of violence and lack of opportunities for certain people due to discrimination, leading to our awareness designs being part of our solution. Leadership analysis had me investigate and present the leadership of U.S president Franklin Delano Roosevelt along with applying leadership models learnt in class.

Other courses included SAT online preparation along with advanced mathematics on introductory calculus material.

SSRN

Independent Researcher

November 2024 - May 2025 (7 months)

Independently designed a project on Venezuelan immigration to the Caribbean island of Trinidad. Interviewed various Venezuelan experts living in Trinidad while conducting research from multiple sources from articles and existing research surrounding the humanitarian crisis. This 42-page publication is ranked against other SSRN publications and also has published in ResearchGate and Academia.

Education

Perkiomen School

· (August 2025 - May 2027)

The International School of Port of Spain

Middle School / High School, Student · (August 2021 - June 2025)